§2 波前

§2.1 波前的概念

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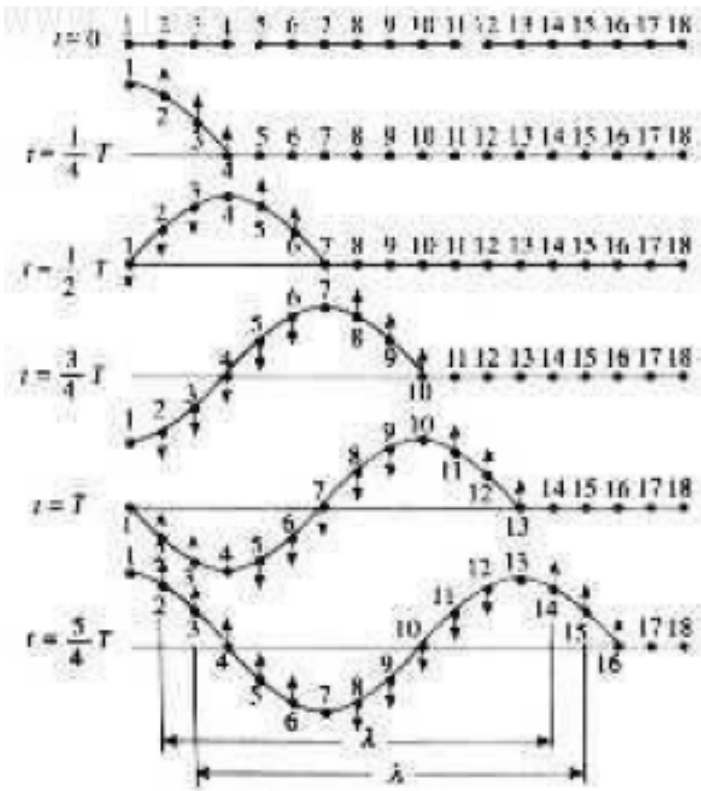
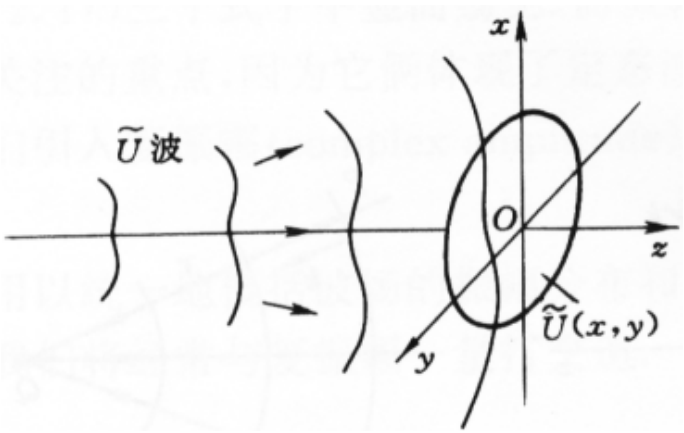
§2.3 傍轴条件和远场条件（轴外物点）

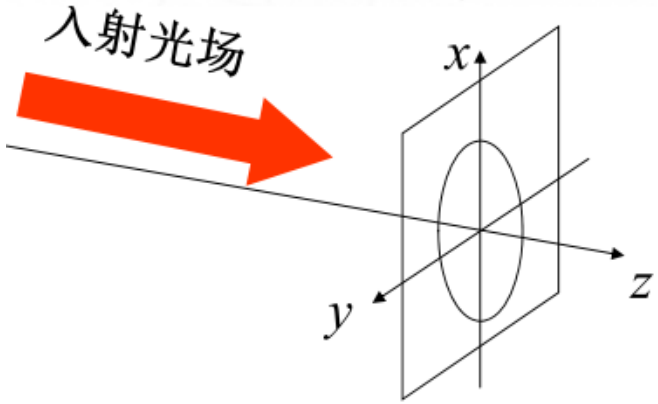
§2.4 高斯光束

2.1 波前的概念



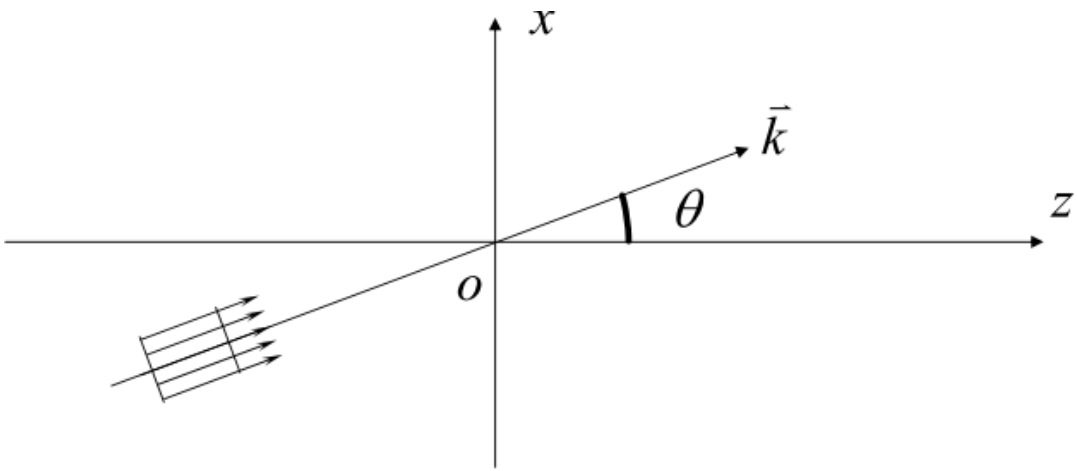
冲击波的波前





波前：波场中的任一曲面。更多地指一个平面，如记录介质、感光底片、接收屏幕等。

例1：平面波的波前。一列平面波，传播方向平行于x-z面，与z轴成倾角θ，求波前z=0面上的复振幅分布



三个波矢分量：$k_x = k\sin\theta$，$k_y = 0, k_z = k\cos\theta$

当φ_0=0时：$\tilde{U}(x,y,z) = Ae^{ik(x\sin\theta + z\cos\theta)}$

在波前 z=0 上：

$$\tilde{U}(x,y) = Ae^{ikx\sin\theta}$$

共轭波：复振幅互为复数共轭的波

例2. 考察例题1中平面波的共轭波

在波前 z=0 上： $\tilde{U}(x,y)=Ae^{ikx\sin\theta}$

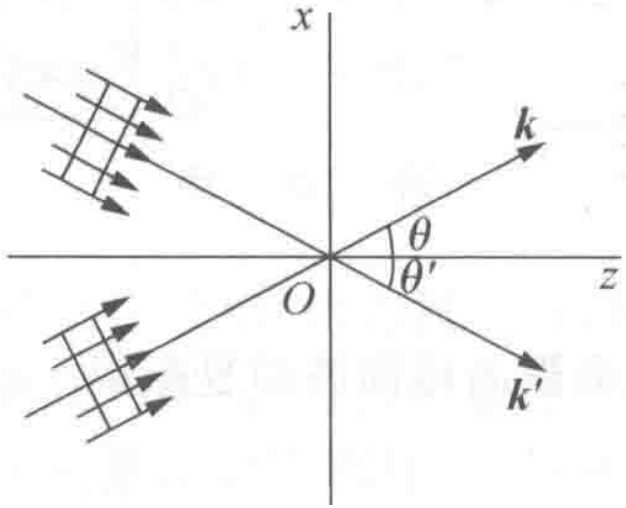
其共轭波为：

$$\tilde{U}^*(x,y)=A\exp[-ikx\sin\theta]=A\exp[ikx\sin(-\theta)]$$

前向共轭波

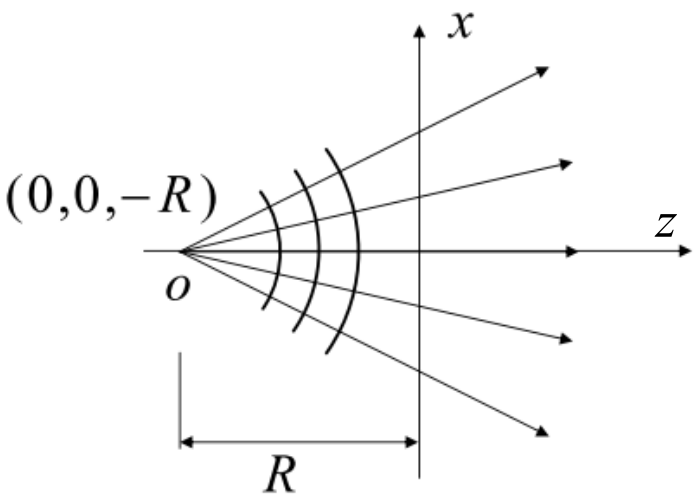
时间反演波

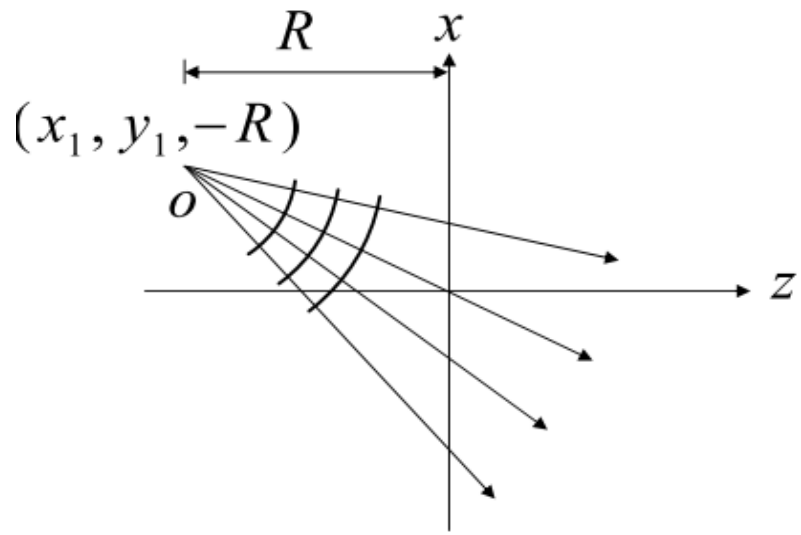
$$\tilde{U}(x,y,z)=Ae^{ik(x\sin\theta+z\cos\theta)}$$



例3：与 $z=0$ 平面距离为R的两个物点在此平面上产生的复振幅分布，一个物点在轴上，另一个物点在轴外

$$\tilde{U}(x,y)=\frac{a}{\sqrt{x^2+y^2+R^2}}e^{ik\sqrt{x^2+y^2+R^2}}$$

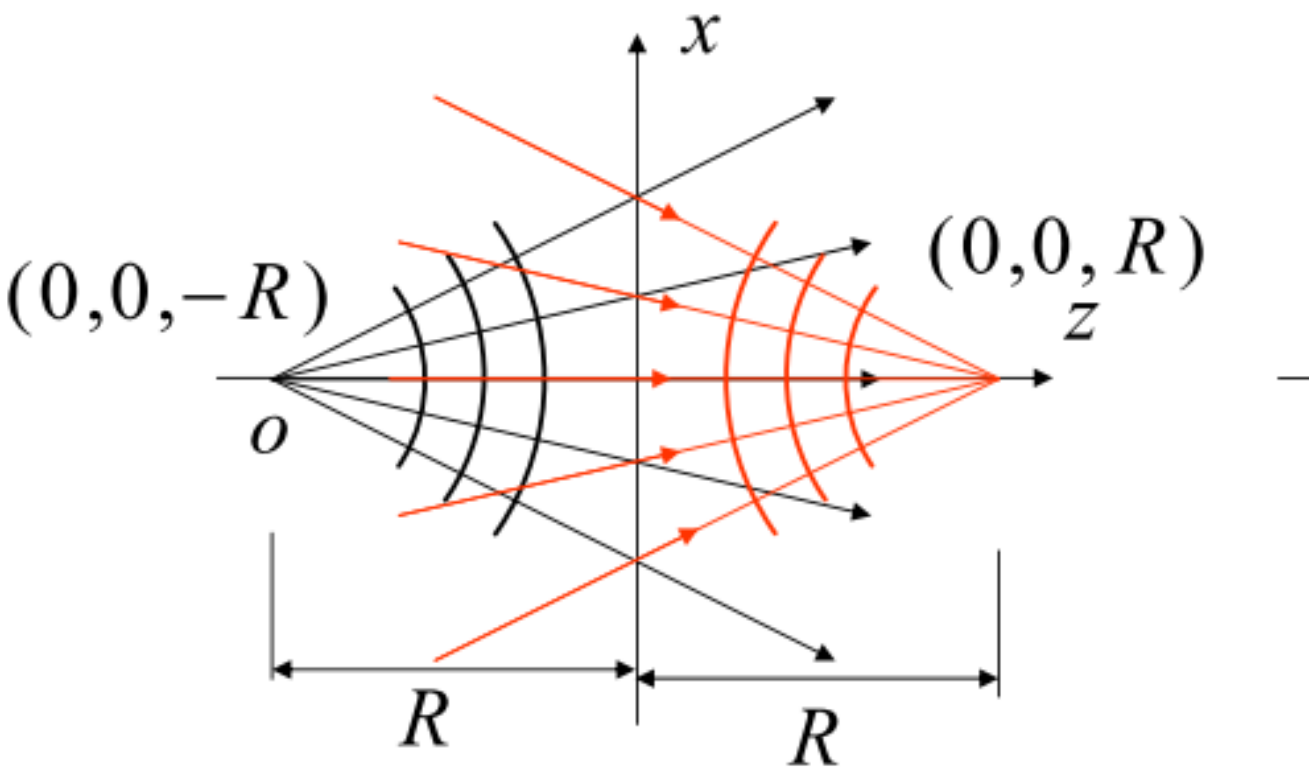




$$\tilde{U}(x,y)=\frac{a}{\sqrt{(x-x_1)^2+(y-y_1)^2+R^2}}e^{ik\sqrt{(x-x_1)^2+(y-y_1)^2+R^2}}$$

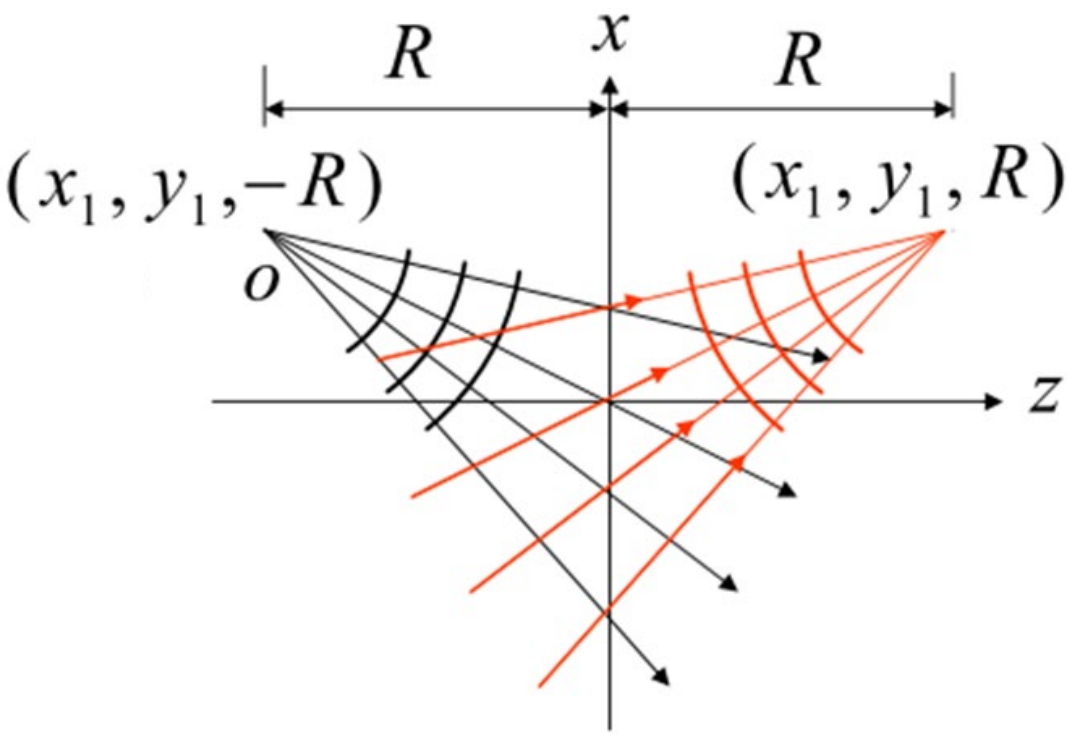
例4. 上述球面波的共轭波

$$\tilde{U}(x,y)=\frac{a}{\sqrt{x^2+y^2+R^2}}e^{-ik\sqrt{x^2+y^2+R^2}}$$



例4. 上述球面波的共轭波

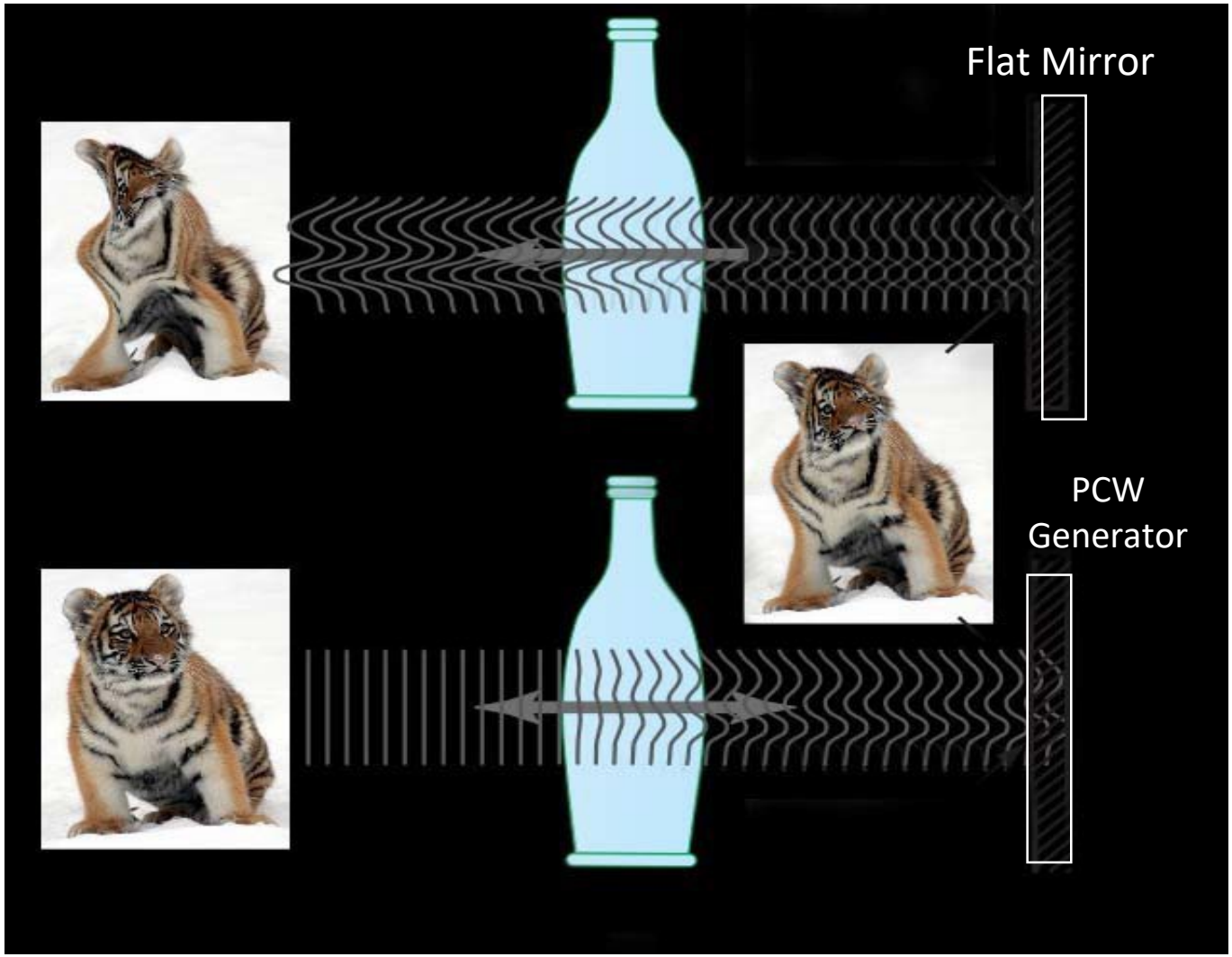
$$\tilde{U}(x,y)=\frac{a}{\sqrt{(x-x_1)^2+(y-y_1)^2+R^2}}e^{-ik\sqrt{(x-x_1)^2+(y-y_1)^2+R^2}}$$



☞ 位相共轭波的产生及其畸变修复功能

普通镜子

共轭镜



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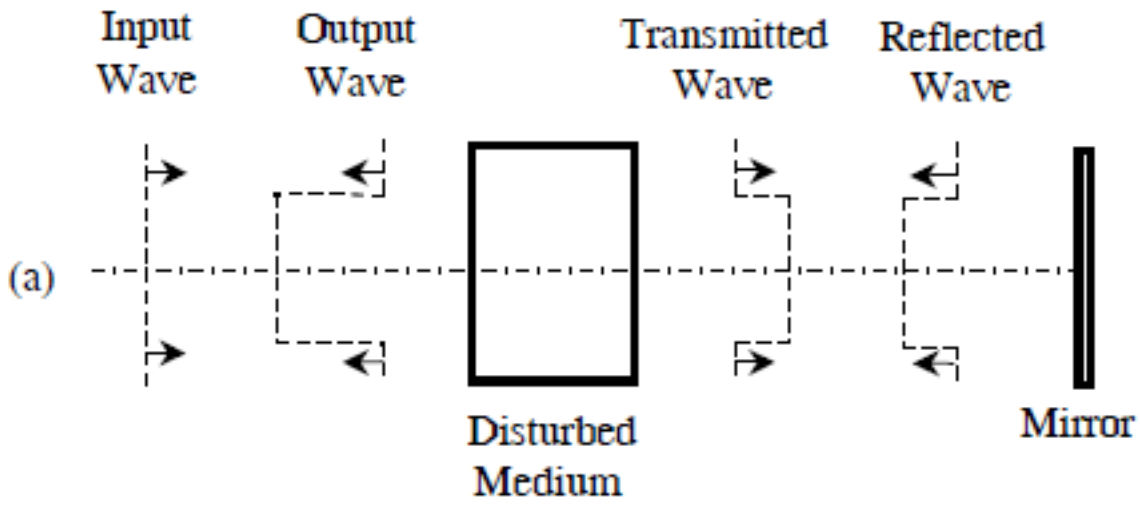
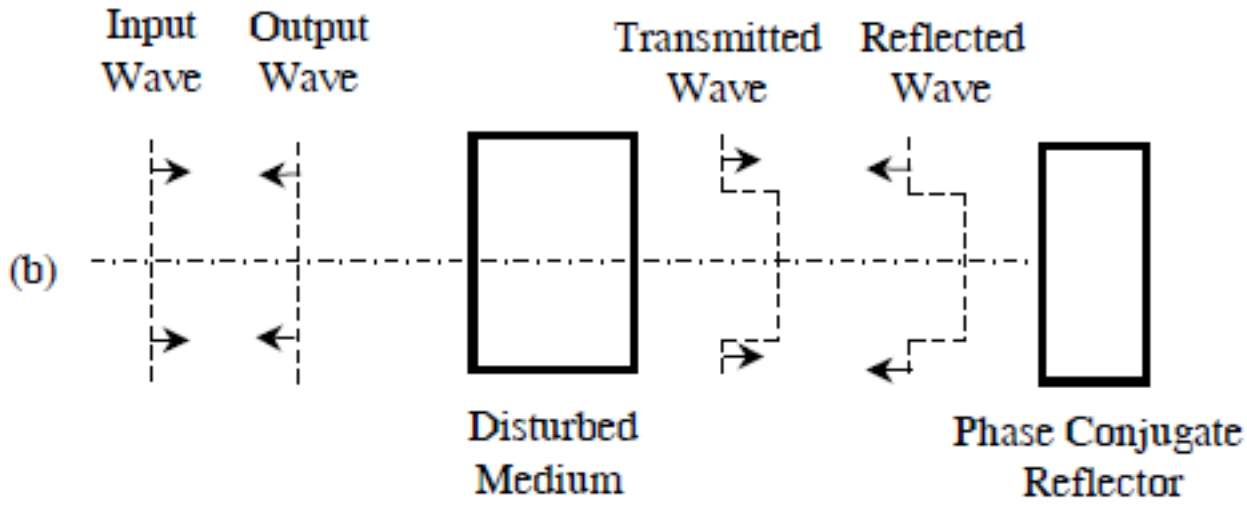
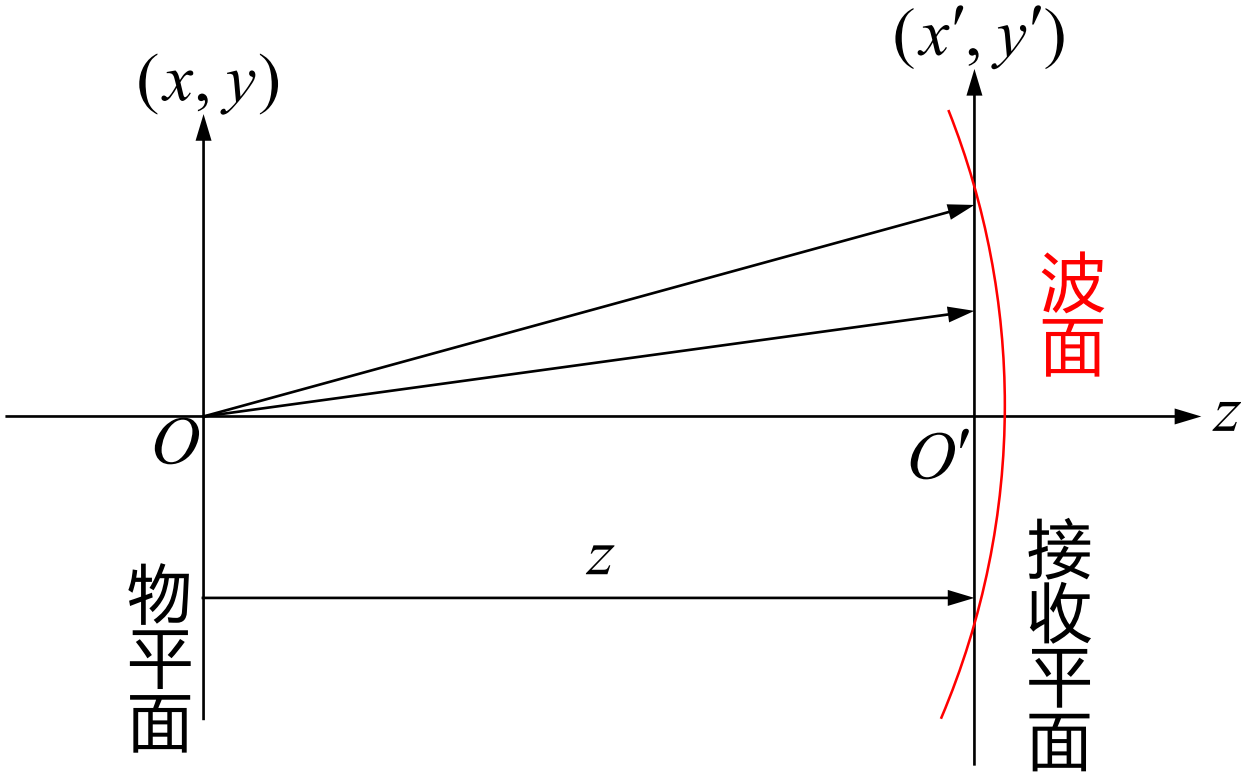


共轭镜



2.2 傍轴(paraxial)条件和远场(far field)条件（轴上物点）

- 接收器通常都是平面屏，光源多是球面波
- 在接收屏上不同位置，相位、振幅都不同



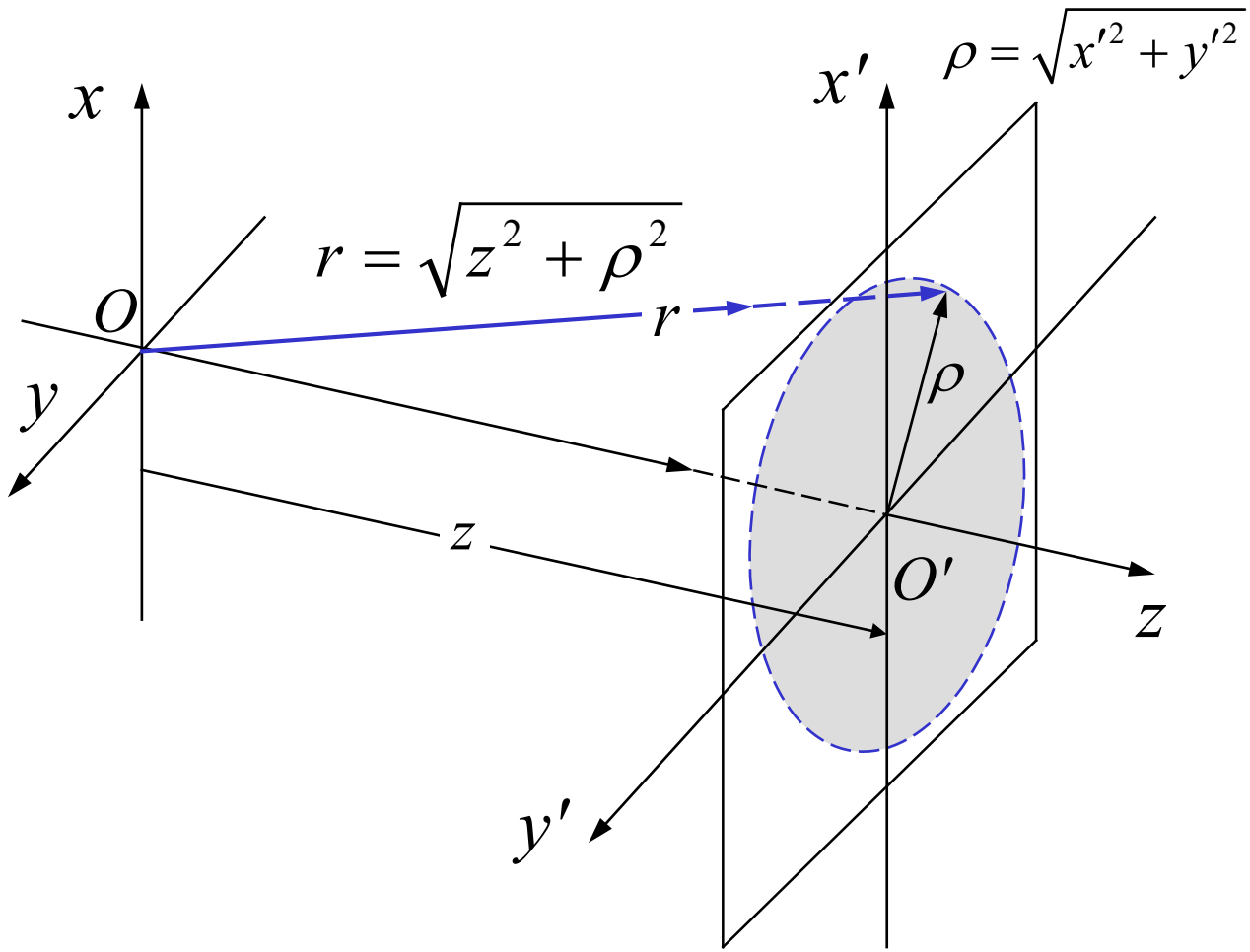
问题：从物理（波动）的意义来考虑，点光源距离与波前线度之比达到什么程度，才能把球面波看做平面波?

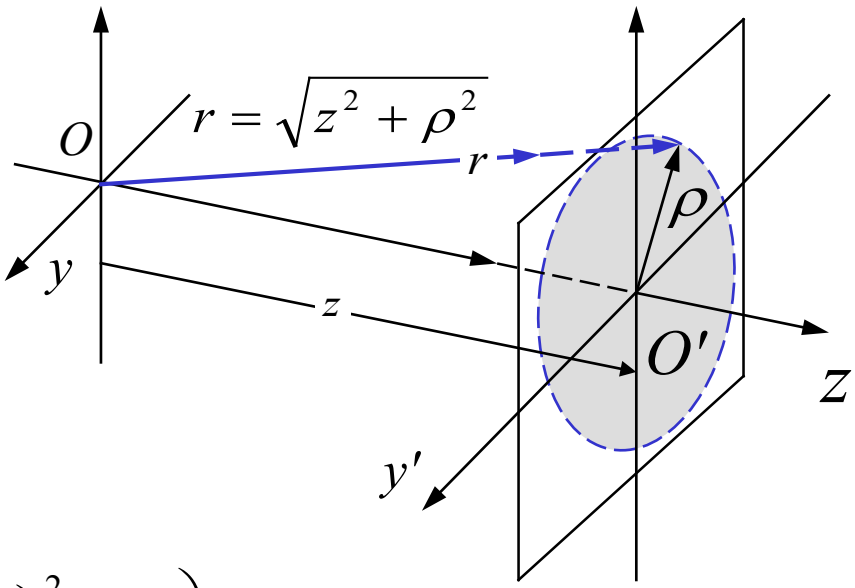
$$A(P)=\frac{a}{\sqrt{z^2+\rho^2}}$$

$$=\frac{a}{z}\frac{1}{\sqrt{1+(\frac{\rho}{z})^2}}$$

$$=\frac{a}{z\left[1+\frac{1}{2}\left(\frac{\rho}{z}\right)^2+\cdots\right]}$$

$$\approx\frac{a}{z\left[1+\frac{1}{2}\left(\frac{\rho}{z}\right)^2\right]}$$





球面波的相位

$$\varphi(P)=\varphi(x',y')=kr=k\sqrt{z^2+\rho^2}$$

$$=kz\sqrt{1+\left(\frac{\rho}{z}\right)^2}=kz\left(1+\frac{\rho^2}{2z^2}+\frac{\rho^2}{2z^2}\left(\frac{\rho}{z}\right)^2+\cdots\right)$$

$$\approx k\left(z+\frac{\rho^2}{2z}\right)$$

$$\tilde{U}(x',y')\approx\frac{a}{z(1+\rho^2/2z^2)}e^{ik(z+\rho^2/2z)}$$

傍轴条件 $\rho^2 / z^2 << 1$ 或 $z^2 >> \rho^2$

$$\tilde{U}(x', y') \approx \frac{a}{z(1+\rho^2/2z^2)} e^{ik(z+\rho^2/2z)}$$
$$\approx \frac{a}{z} e^{ik(z+\rho^2/2z)}$$

远场条件 $k\rho^2/2z << \pi$ 或 $z >> \rho^2 / \lambda$

$$\tilde{U}(x', y') \approx \frac{a}{z(1+\rho^2/2z^2)} e^{ik(z+\rho^2/2z)}$$
$$\approx \frac{a}{z(1+\rho^2/2z^2)} e^{ikz}$$

傍轴条件和远场条件的关系

两条件相互独立，究竟哪个的限制更强与具体情况（波长）有关。在光波波段，往往是远场条件蕴含傍轴条件。

当两条件同时满足时：

$$\tilde{U}(x', y') \approx \frac{a}{z} e^{ikz}$$

即为正入射的平面波。

傍轴条件：波前上的振幅是常数

远场条件：波前的相位是直角坐标系的线性函数

例5

λ~0.5μm，ρ~1mm，估算满足傍轴条件、远场条件的距离

取>>为50倍：

傍轴：$z_1^2 = 50\rho^2$

$z_1 = \sqrt{50}\rho \approx 7mm$

远场：$z_2 = 50\rho^2/\lambda \approx 100\,m$

远场蕴含傍轴

例5

λ~0.5μm，ρ~1mm，估算满足傍轴条件、远场条件的距离

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远场蕴含傍轴

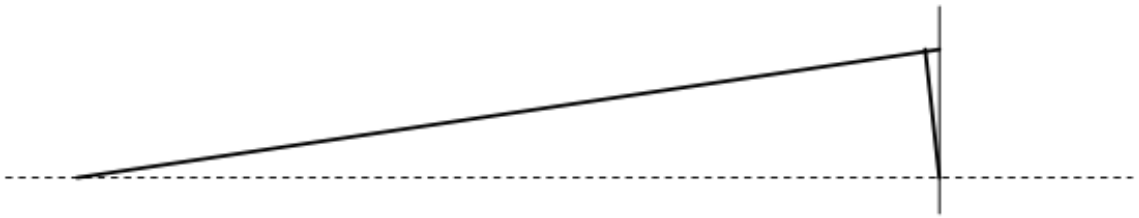
远场：$z_2 = 50\rho^2/\lambda \approx 100\,m$

λ~1m(声波)，ρ~10cm，估算满足傍轴条件、远场条件的距离

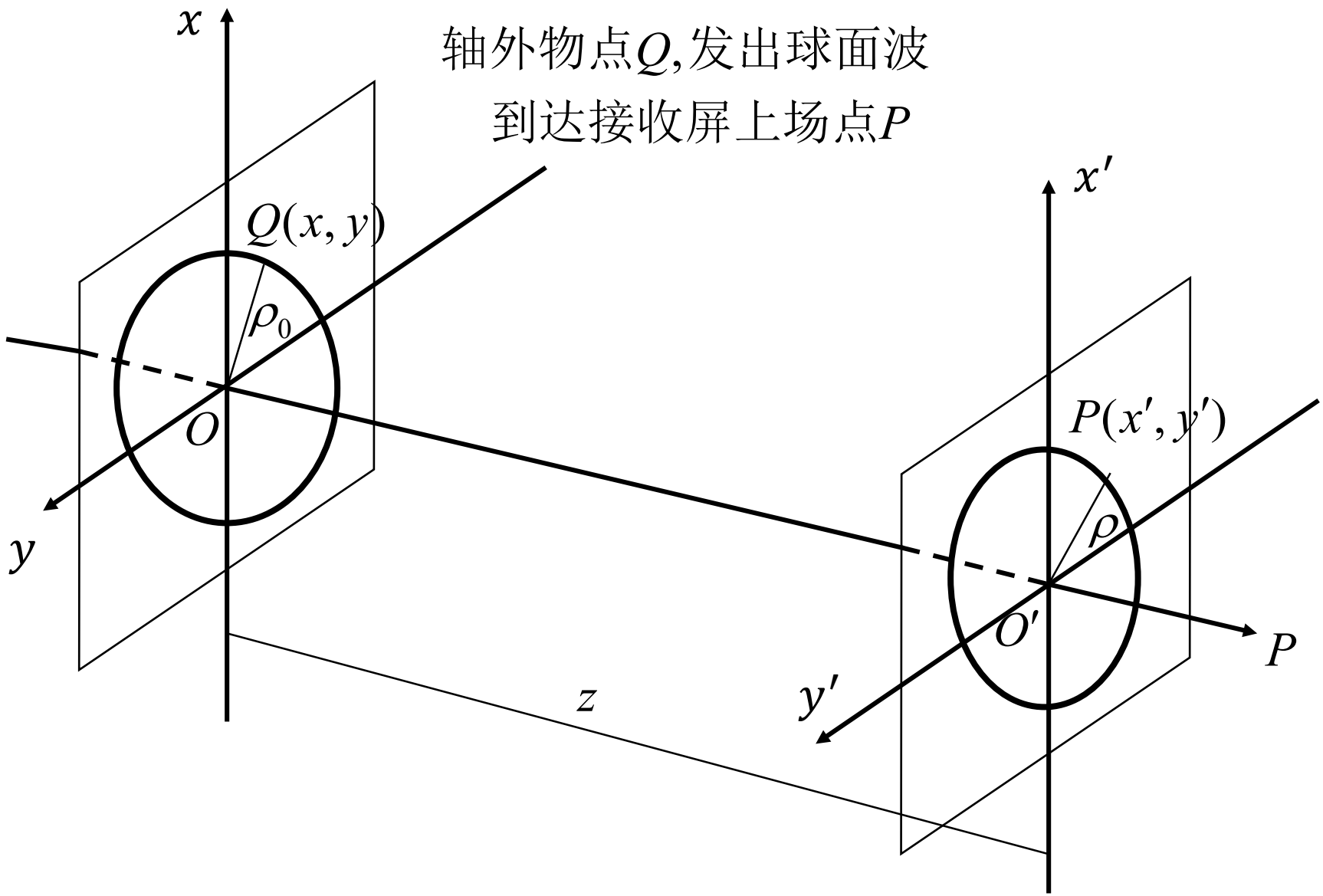
傍轴：$z_1 = \sqrt{50}\rho \approx 70\,cm$

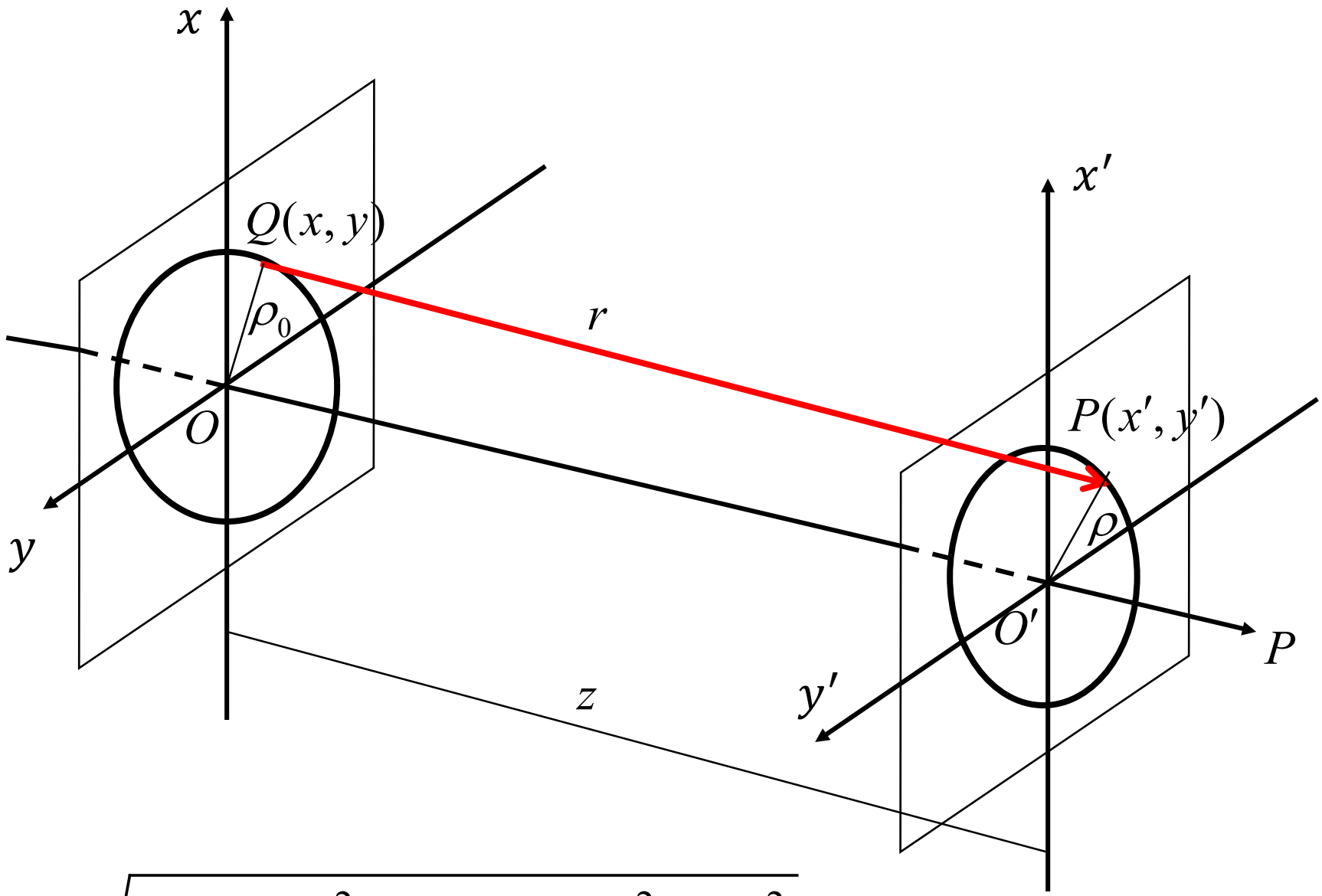
傍轴蕴含远场

远场：$z_2 = 50\rho^2/\lambda \approx 50\,cm$

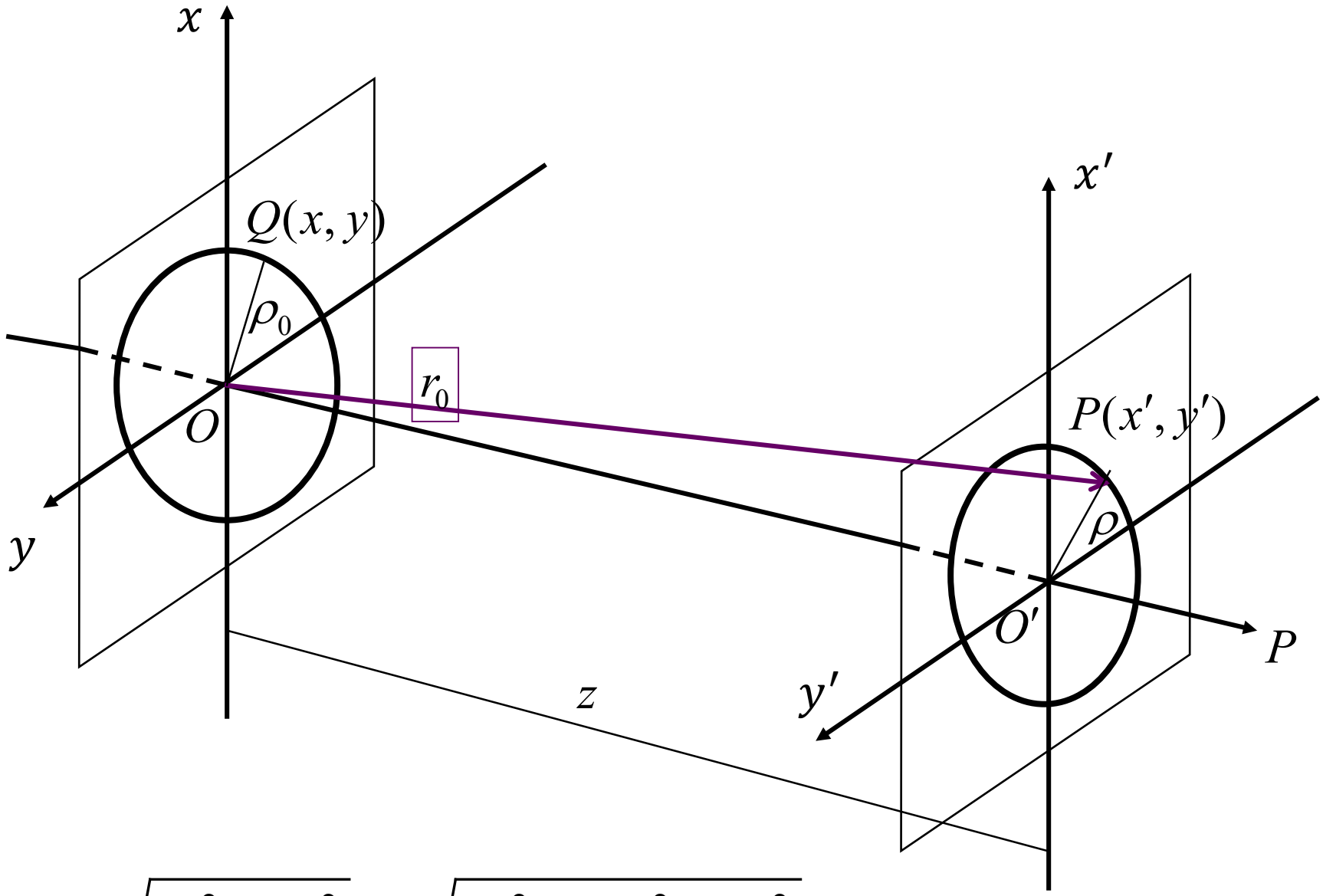


2.3 傍轴条件和远场条件（**轴外物点**）

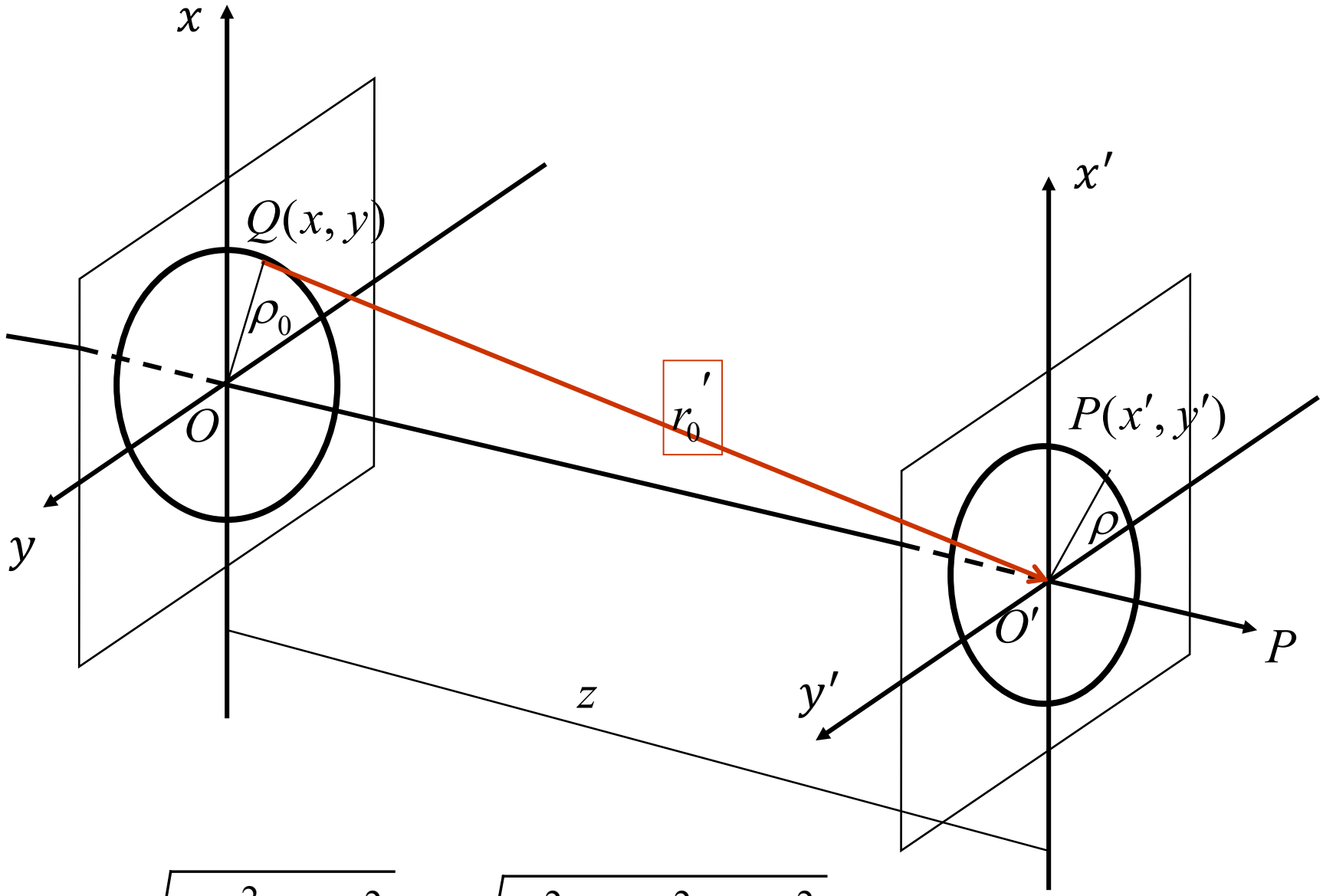




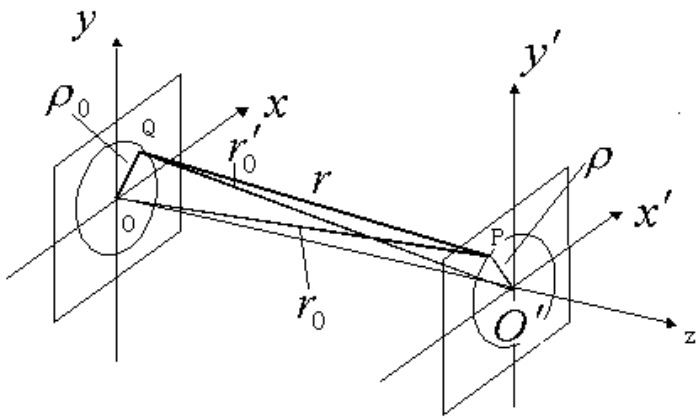
$$r = \sqrt{(x - x')^2 + (y - y')^2 + z^2}$$



$$r_0 = \sqrt{\rho^2 + z^2} = \sqrt{x'^2 + y'^2 + z^2}$$



$$r_0' = \sqrt{\rho_0{}^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$$



$$r = \sqrt{(x - x')^2 + (y - y')^2 + z^2}$$

$$r_0 = \sqrt{\rho^2 + z^2} = \sqrt{x'^2 + y'^2 + z^2}$$

$$r_0' = \sqrt{\rho_0{}^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$$

Q到O的距离 $\rho_0 = \sqrt{x^2 + y^2}$

P到O′ 的距离 $\rho' = \sqrt{x'^2 + y'^2}$

对上述距离做泰勒展开，有

$$r_0 \approx z + \frac{x'^2 + y'^2}{2z} \qquad r'_0 \approx z + \frac{x^2 + y^2}{2z}$$

$$r = \sqrt{(x - x')^2 + (y - y')^2 + z^2}$$

$$= z\sqrt{\frac{(x - x')^2}{z^2} + \frac{(y - y')^2}{z^2} + 1}$$

$$\approx z + \frac{x'^2 + y'^2}{2z} + \frac{x^2 + y^2}{2z} - \frac{xx' + yy'}{z}$$

$$= r'_0 + \frac{x'^2 + y'^2}{2z} - \frac{xx' + yy'}{z}$$

$$\boxed{r'_0 \approx z + \frac{x^2 + y^2}{2z}}$$

或者 $$= r_0 + \frac{x^2 + y^2}{2z} - \frac{xx' + yy'}{z}$$

$$\boxed{r_0 \approx z + \frac{x'^2 + y'^2}{2z}}$$

点源 Q 在 $x'-y'$ 面上激发的球面波前为

$$\tilde{U}(x',y')=\frac{a}{r}\exp[ikr]$$

$$=\frac{a}{\sqrt{(x-x')^2+(y-y')^2+z^2}}\exp[ik\sqrt{(x-x')^2+(y-y')^2+z^2}]$$

(1) 物点、场点同时满足傍轴条件： $x^2, y^2, x'^2, y'^2 << z^2$

$$\tilde{U}(x',y')\approx\frac{a}{z}e^{ik(r_0+\frac{x^2+y^2}{2z})}e^{-i\frac{k}{z}(xx'+yy')}$$

$$\approx\frac{a}{z}e^{ik(r_0'+\frac{x'^2+y'^2}{2z})}e^{-i\frac{k}{z}(xx'+yy')}$$

（2）场点满足傍轴条件，物点同时满足傍轴和远场条件：

场点P满足傍轴条件： $x'^2, y'^2 << z^2$

物点Q同时满足

傍轴条件： $x^2, y^2 << z^2$

远场条件： $\frac{x^2}{\lambda}, \quad \frac{y^2}{\lambda} << z$

$$\tilde{U}(x', y') \approx \frac{a}{z} e^{ik\left(r_0 + \frac{x^2+y^2}{2z}\right)} e^{-i\frac{k}{z}(xx'+yy')}$$

$$\approx \frac{a}{z} e^{ikr_0} e^{-i\frac{k}{z}(xx'+yy')}$$

平面波？？

(3) 物点满足傍轴条件，场点同时满足傍轴和远场条件：

物点Q满足傍轴条件：$x^2, y^2 << z^2$

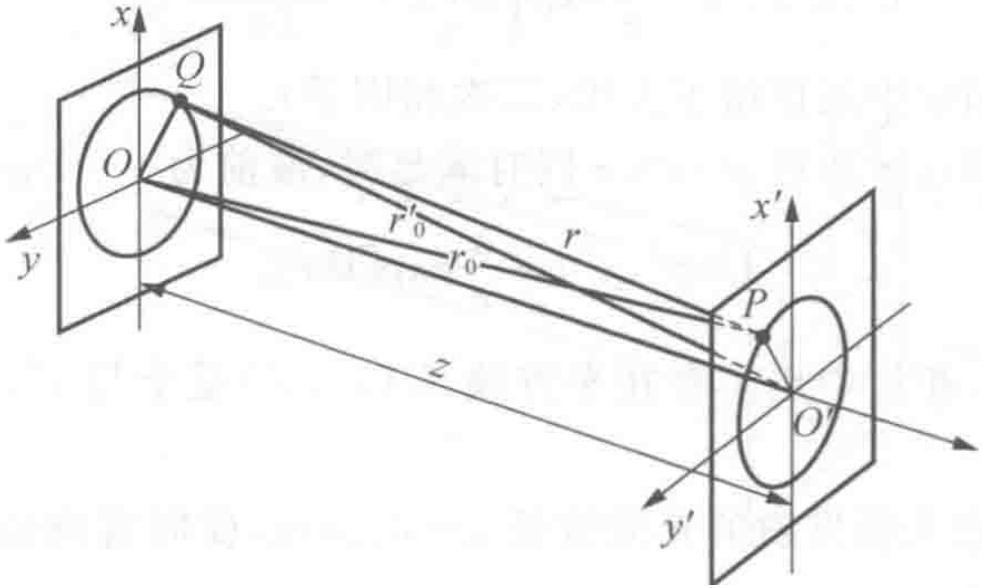
场点P满足：

傍轴条件： $x'^2, y'^2 << z^2$

远场条件： $\frac{x'^2}{\lambda}, \frac{y'^2}{\lambda} << z$

$$\tilde{U}(x', y') \approx \frac{a}{z} e^{ikr_0'} e^{-i\frac{k}{z}(xx'+yy')}$$

位相是x', y'的线性函数 → 平面波



平面波 $$\tilde{U}(x',y') \approx \frac{a}{z} e^{ikr'_0} e^{-i\frac{k}{z}(xx'+yy')}$$

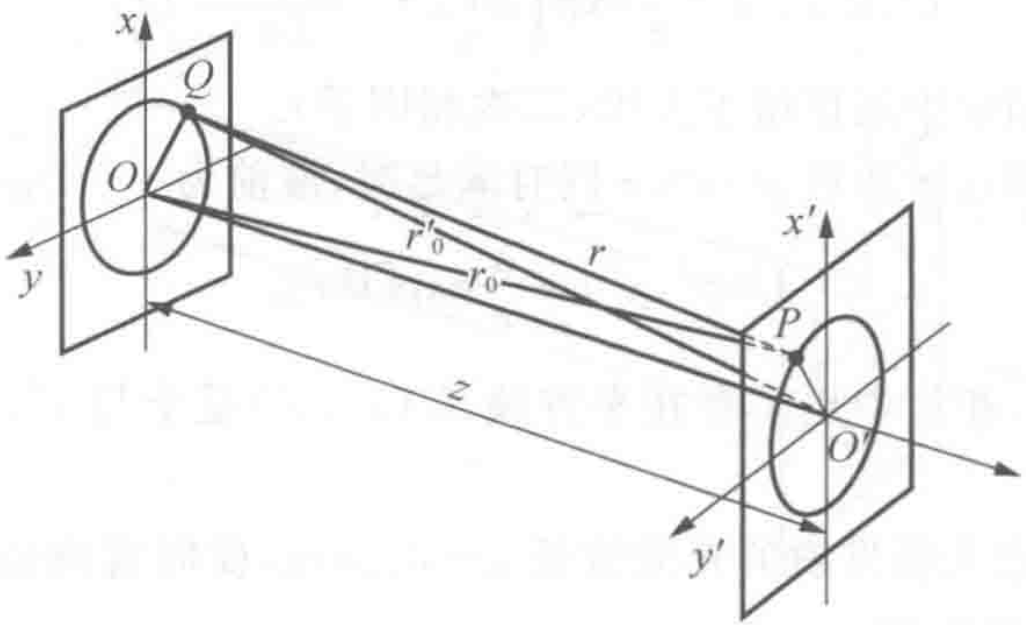
接收平面上一斜入射的平面波，波矢方向是物点到接收平面中心的连线，其方向余弦是：

$$\cos\alpha' = -\frac{x}{z}, \quad \cos\beta' = -\frac{y}{z}$$

沿QO′ 方向。

傍轴条件、远场条件可看着球面波向平面波的转化。

输入端



输出端

实空间 ⟷ 频谱空间

对于场点$P(x',y')$:

$$(x,y) \longleftrightarrow -\frac{k}{z}(x,y)$$

对于源点Q(x,y):

$$(x',y') \longleftrightarrow -\frac{k}{z}(x',y')$$

（4）物点和场点均同时满足傍轴条件和远场条件：

物点Q满足傍轴条件和远场条件：$x^2, y^2 << z^2$　　$\frac{x^2}{\lambda}, \frac{y^2}{\lambda} << z$

场点P满足傍轴条件和远场条件：$x'^2, y'^2 << z^2$　　$\frac{x'^2}{\lambda}, \frac{y'^2}{\lambda} << z$

$$\tilde{U}(x', y') = \frac{a}{r}\exp[ikr]$$

$$= \frac{a}{\sqrt{(x-x')^2+(y-y')^2+z^2}}\exp[ik\sqrt{(x-x')^2+(y-y')^2+z^2}]$$

$$r = \sqrt{(x-x')^2+(y-y')^2+z^2}$$

$$\approx z + \frac{x'^2+y'^2}{2z} + \frac{x^2+y^2}{2z} - \frac{xx'+yy'}{z}$$

$$\tilde{U}(x', y') = \frac{a}{z}exp(ikz)$$

近似可以看成是平行于z轴的平面波

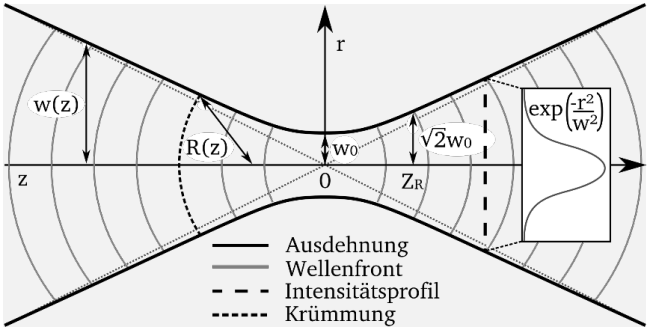
2.4 高斯光束（Gaussian beam）

光学谐振腔(optical resonant cavity)内能够稳定存在的一种光场，其复振幅描述为：

$$\tilde{U}(x,y,z)=\frac{A}{\omega(z)}e^{-\frac{x^2+y^2}{\omega^2(z)}}e^{-ik\left[\frac{x^2+y^2}{2r(z)}+z\right]+i\varphi(z)}$$

其中： $\omega(z)=\omega_0\left(1+\frac{\lambda^2 z^2}{\pi\omega_0^4}\right)^{1/2}$

光束有效半径



$$r(z)=z\left(1+\frac{\pi\omega_0^4}{\lambda^2 z^2}\right)$$

$\omega=\omega_0$ 时光束最细

ω_0 ： **束腰(beam waist)**位置取决于谐振腔的特性

$$\tilde{U}(x,y,z)=\frac{A}{w(z)}\exp\left[-\frac{x^2+y^2}{w^2(z)}\right]\exp\left[-ik\left(\frac{x^2+y^2}{2r(z)}+z\right)+i\varphi(z)\right]$$

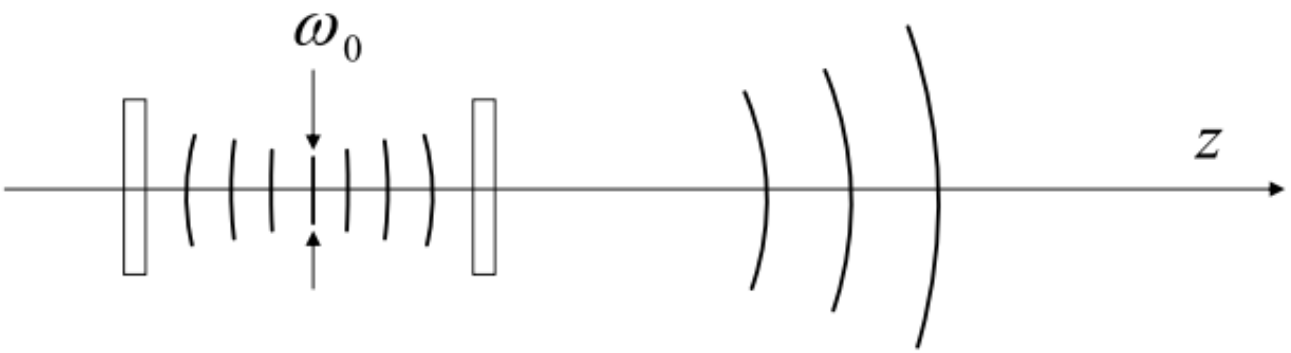
$$\begin{cases} w(z)=w_0\left(1+\dfrac{\lambda^2 z^2}{\pi w_0^4}\right)^{1/2} \\ r(z)=z\left(1+\dfrac{\pi^2 w_0^4}{\lambda^2 z^2}\right). \end{cases}$$

一般情况下

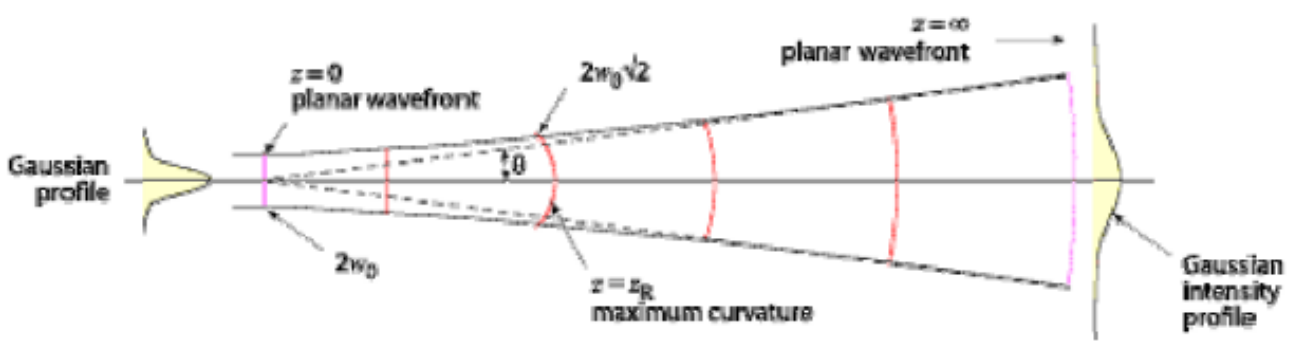
$r(z)\neq z$

远场范围内

$z^2\lambda^2 \gg \pi^2 w_0^4,\ r(z)\approx z$



束腰处z=0
平面波特征

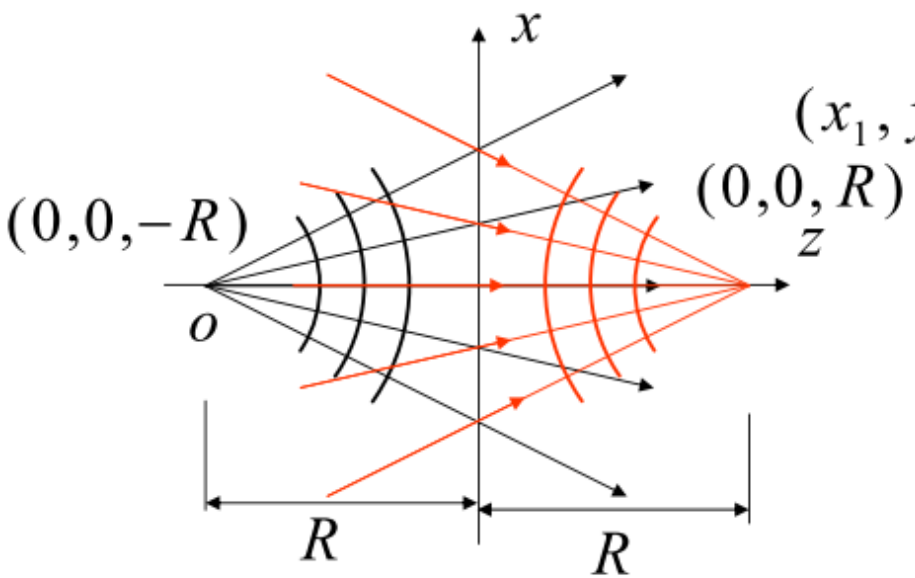


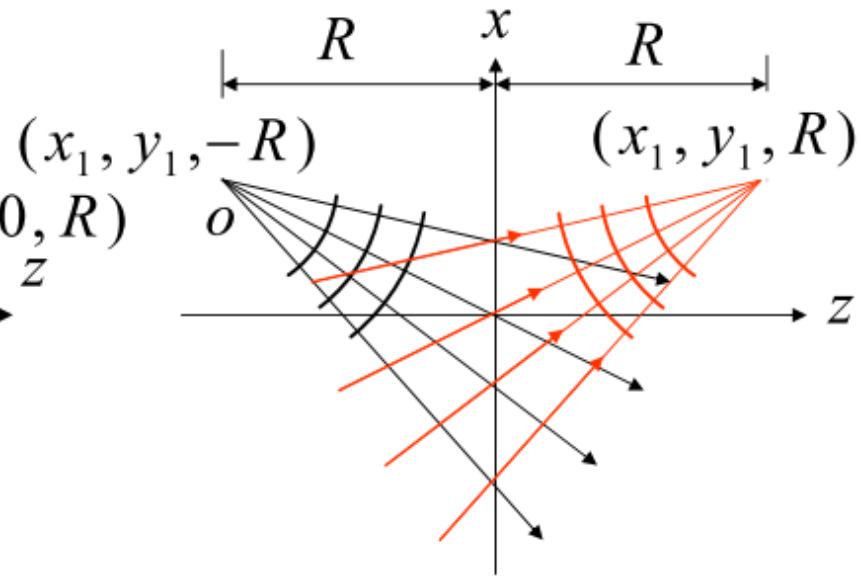
球心在不同位置处的球面波

作业：P117 --1, 2

例4. 上述球面波的共轭波

$$\tilde{U}(x,y)=\frac{a}{\sqrt{x^2+y^2+R^2}}e^{-ik\sqrt{x^2+y^2+R^2}}$$





$$\tilde{U}(x,y)=\frac{a}{\sqrt{(x-x_1)^2+(y-y_1)^2+R^2}}e^{-ik\sqrt{(x-x_1)^2+(y-y_1)^2+R^2}}$$